Claude Code and OpenCode: A Crash Course

General Agents, Chapter 3/12 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/agentic-coding

EN: The whole course is one idea: context engineering -- deliberately deciding what the model sees. 3 things to remember from Karpathy: it's a ghost not a human, its intelligence is jagged, trust it only as much as you can verify.
Roman Urdu: Poora course ek idea ke ird gird hai: context engineering -- jaan-boojh kar decide karna model kya dekhe. Karpathy se 3 baatein: yeh ghost hai insaan nahi, intelligence jagged hai, utna hi trust karo jitna verify ho sake.

1. Foundations

- Chatbots answer; Claude Code/OpenCode TAKE ACTIONS -- biggest shift is stop asking questions, start giving instructions.
- Plan Mode = 'look but do not touch' (rule of thumb: tasks over 10 min). Permissions discipline: watch everything at first, then auto-approve safe actions.
- 'Strong model to plan, cheap model to execute.'

2. Context Management

- Context is a stack of layers, each costing tokens: system prompt+tools (fixed), rules file (fixed), conversation (growing), referenced files (on-demand), active skill (on-demand).
- **Context rot** is real -- rules get ignored, finished work redone. Rule: compact or clear once you've used half the window.
- **Doom loop signs:** saying sorry without fixing, repeating the same code, mentioning files that don't exist -- stop messaging, reset instead.

3. Rules File + Skills/Hooks/Subagents

- CLAUDE.md/AGENTS.md is read at the start of EVERY conversation. Test every line: 'if I deleted this, would AI make a mistake?' Prune contradictions.
- 3 distinct tools: Command/Skill (saved reusable prompt), Hook/Plugin (always-runs rule), Subagent (isolated-window helper).
- **Verification Loop** -- the book's single most important pattern: Attempt -> Check -> Fix -> Repeat, no human in between.

4. MCP + Composing Tools

- MCP connects AI to real services (Slack, GitHub, DBs) -- a standard wrapper over APIs. Start with 1-2 connections.
- Prompt injection: fetched content should only inform, never instruct -- treat strange behavior as suspicious.
- Git worktrees let parallel sessions run without file conflicts.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Context engineering	Deliberately deciding what the model sees.	Jaan-boojh kar decide karna model kya dekhe.
Context rot	Quality degradation as a conversation fills up with unmanaged context.	Conversation context se bharne se quality degrade hona.
Verification Loop	Attempt -> Check -> Fix -> Repeat, with no human in the middle.	Attempt -> Check -> Fix -> Repeat, beech mein insaan nahi.
CLAUDE.md / AGENTS.md	The rules file read at the start of every conversation.	Rules file jo har conversation ke shuru mein parhi jati hai.
Subagent	An isolated-context helper that returns only a summary.	Isolated-context helper jo sirf summary wapas deta.
MCP	The standard connecting an agent to external services like Slack or GitHub.	Standard jo agent ko Slack/GitHub jaisi services se connect karta.

Quick Self-Check

Question	Answer
What's the single biggest shift when moving from chatbot to coding agent?	Stop asking questions, start giving instructions -- the agent takes action instead of just answering.
What's the fix when you spot 'doom loop' signs (apologizing without fixing, repeated code)?	Stop sending messages and reset with /compact or /clear instead of continuing to prompt.
What's the test for whether a rules-file line should stay?	'If I deleted this line, would the AI make a mistake?' -- if no, delete it.
Why does a subagent return only a summary, not its full transcript?	To keep the main context window clean -- but that means a confident wrong claim inside its work can hide inside that summary.
What's the safe default for handling fetched/untrusted content?	It should only inform the agent, never instruct it -- treat it as data, not commands.

Where This Leads / Sources

Leads to: Spec-Driven Development, The Four Layers, Harness Engineering. Full worked 8-step example in 05-complete-worked-example.md.

Source: docs/agentic-coding/ (this repo), content originally via Zia Tutor AI connector.